AI FOR GREENER FACTORIES

TerraVow

Healthy machines.
Cleaner factories.
Growth that lasts.

AI that links machine health to pollution, tells operators what to fix and when, and shows greener alternatives with the incentives that can pay for them.

Gemini | Cloud Run | Firestore



[Team name] | [Hackathon and theme label] | [Date]

Factories grow the world, and they are among its biggest polluters

THE GROWTH ENGINE

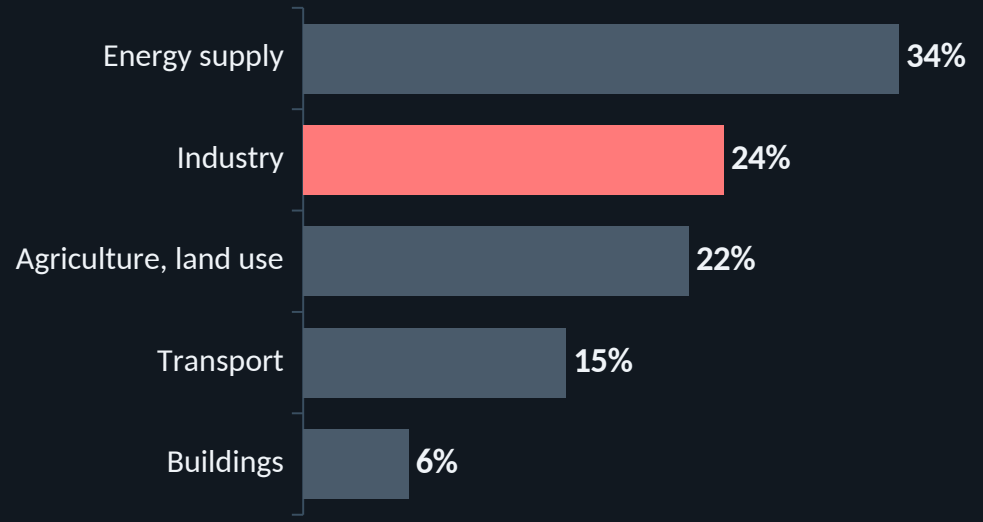
~15% of world GDP is manufacturing value added

16-17% of India's GDP is manufacturing

35.4% of India's manufacturing output comes from MSMEs

THE COST

Share of 2019 global GHG emissions, by sector

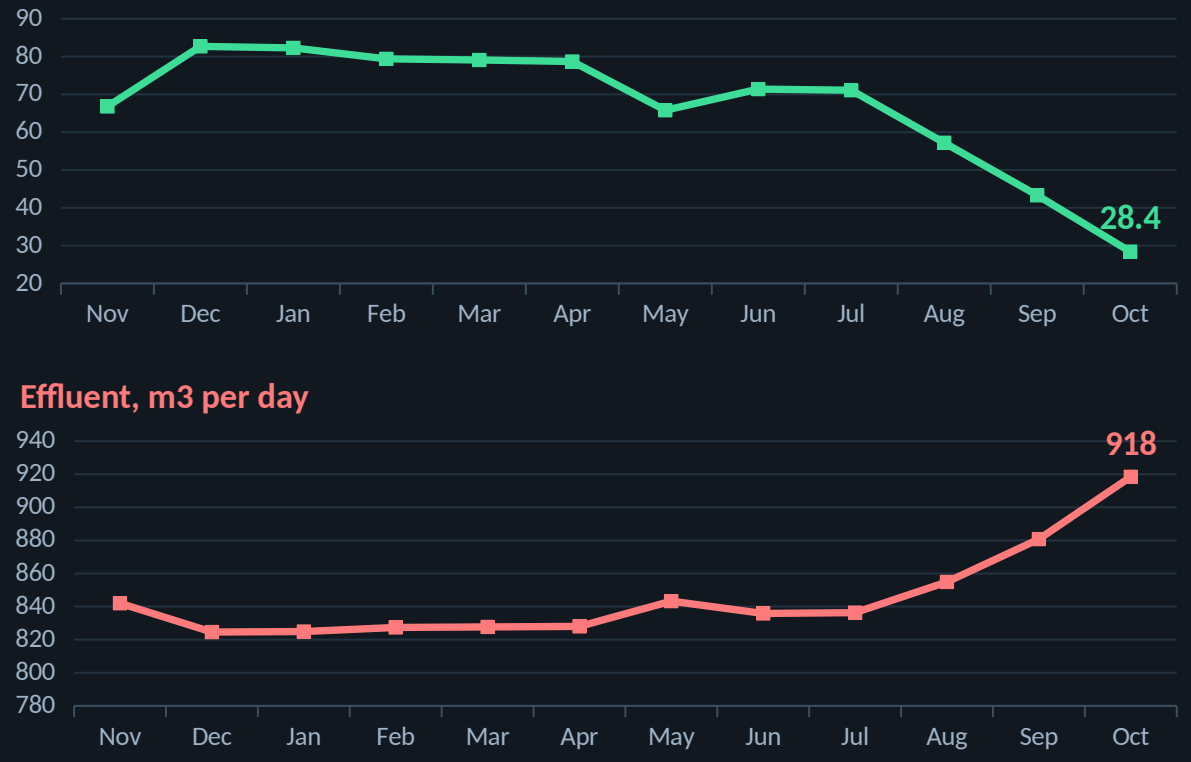


Industry: 24% direct, 34% once electricity is attributed to the factories that use it.

Sources: World Bank NV.IND.MANF.ZS (2025); IPCC AR6 WGIII SPM B.2.1 (2019); PIB 14 Oct 2026 and 14 May 2026. Manufacturing value added and the IPCC industry sector are different scopes.

The hidden cause: worn machines quietly burn more and leak more

KILN K-4 OVER 12 MONTHS | ILLUSTRATIVE



SMALL SNAGS, BIG CONSEQUENCES

A worn seal — Air leaks in. The burner works harder.

A drifting burner — Soot builds up. Fuel and NOx rise.

A clogged filter — The fan strains. Emissions per tonne climb.

5-20% lower energy bills with good O&M

US DOE estimate for federal sites. Direction, not a promise for any one plant.

Left: Kiln K-4, synthetic demo fleet, illustrative engine model (not a measurement). Right: US DOE FEMP O&M Best Practices Guide, Release 3.0 (US federal sites).

Today's tools cannot connect a deferred repair to a pollution event

Maintenance systems

CMMS / EAM, predictive maintenance

WHAT IT DOES

Tracks downtime and work orders

BLIND SPOT

Blind to emissions cost

ESG and carbon accounting

Scope 1/2/3 ledgers

WHAT IT DOES

Reports what was already emitted

BLIND SPOT

Industry averages, not your machine

Compliance and monitoring

EHS workflow, CEMS

WHAT IT DOES

Above or below the limit, right now

BLIND SPOT

Not why, and often self-reported

Plain LLM chatbot

Ask a general model

WHAT IT DOES

Fluent, quick answers

BLIND SPOT

Numbers you cannot audit

Nobody connects **"we deferred this repair"** to **"we just created avoidable emissions and cost"**

And factories cannot just shut down. They need a practical path that keeps output growing while they get cleaner.

Category-level assessment from docs/competitive-landscape.md and docs/overview.md. Individual vendors may differ.

TerraVow is the bridge: from machine health to cleaner, funded upgrades

1 **Report**

Snags logged per machine model

→

2 **Health**

Score and remaining life

→

3 **Pollution**

Air, water, chemical, soil

→

4 **Greener options**

Machines and process changes

→

5 **Funded payback**

India incentives, MSME or large

Safety gate

Vetoes any option that raises total emissions or merely moves pollution from air to water.

Audit trail

Every recommendation and closure is written to a hash-chained log.

Numbers you can trust

Gemini reasons and writes. A deterministic engine computes every figure.

Prototype scope: Requirements 18-30 (Python/FastAPI). See roadmap for what is not built.

One flow, four industries, each with its own AI-built template

Textile dyeing & finishing

AIR WATER CHEMICAL

SAMPLE MODELS

Bosch Industrial UL-S 6000

Thermax Heatpac TH-1100

Thies iMaster

Metals: steel & foundry

AIR WATER SOIL

SAMPLE MODELS

Tenova Walking-Beam Reheat Furnace WB-200

Inductotherm Power-Trak 5000

Danieli Digimelter

Chemicals & fertiliser

AIR WATER CHEMICAL SOIL

SAMPLE MODELS

GEA Niro SD-25

Alfa Laval NX 4000

thyssenkrupp Granulation Drum GD-4

Mining & mineral processing

AIR WATER SOIL

SAMPLE MODELS

FLSmidth Rotary Kiln RK-4.0

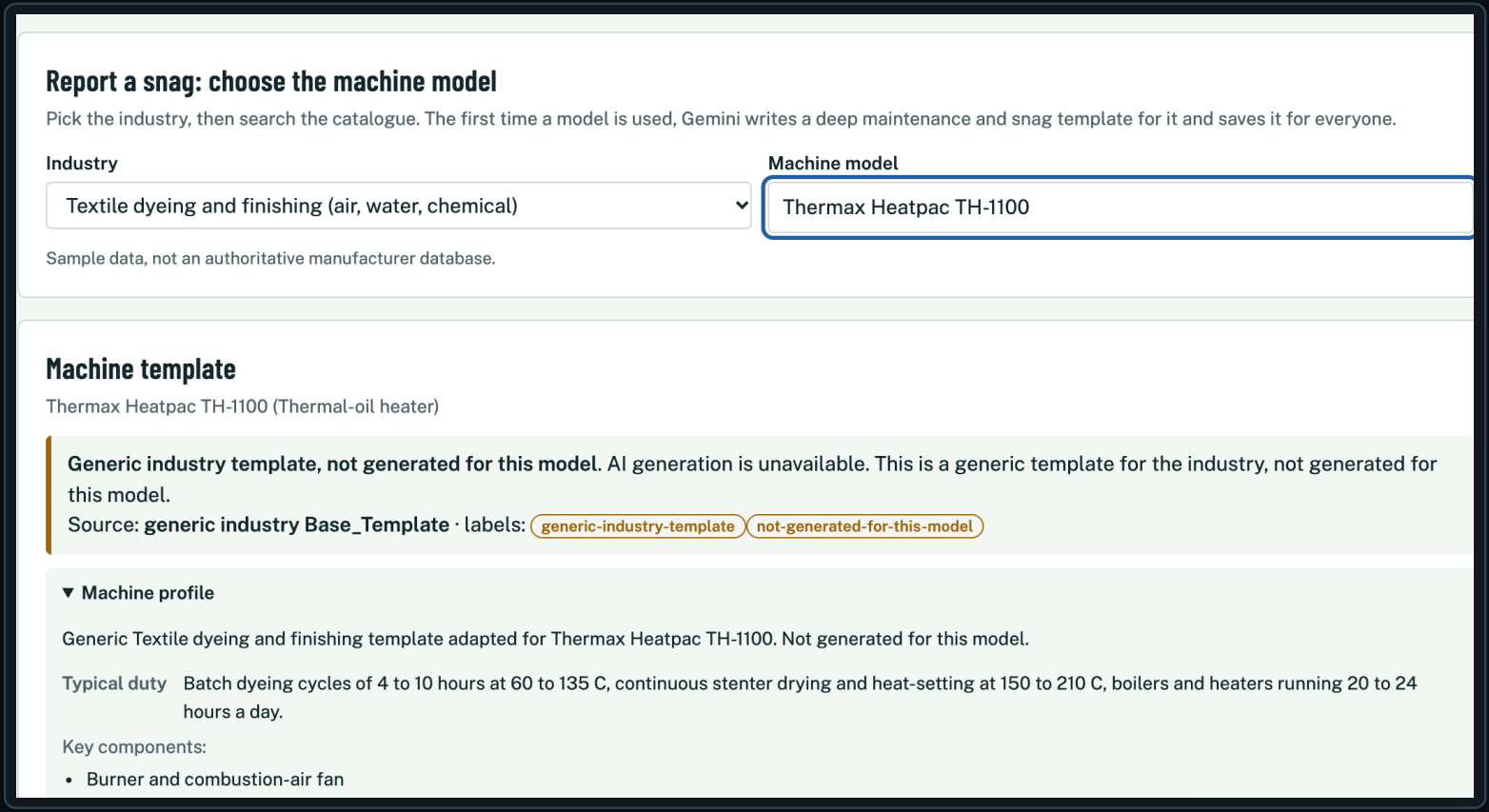
Caterpillar G3516

Metso Nordberg HP5

On India's regulator list: CPCB flags 17 highly polluting categories, including dyes, fertiliser, iron and steel and cement. 4,578 such units reported (Dec 2022).

Domains from prototype/data/industries.json. Catalogue is sample data. CPCB: Rajya Sabha Starred Q101, 15 Dec 2022 (sansad.in).

Step 1 | Report: pick a machine, get a form built for that exact model



1 Search or add

Autocomplete over 20 sample models. Not listed? Add it; Gemini with Google Search looks up specs.

2 Form built for the model

First use triggers a Gemini template: 6+ job types, 12+ snag types, 8+ parameters. Cached.

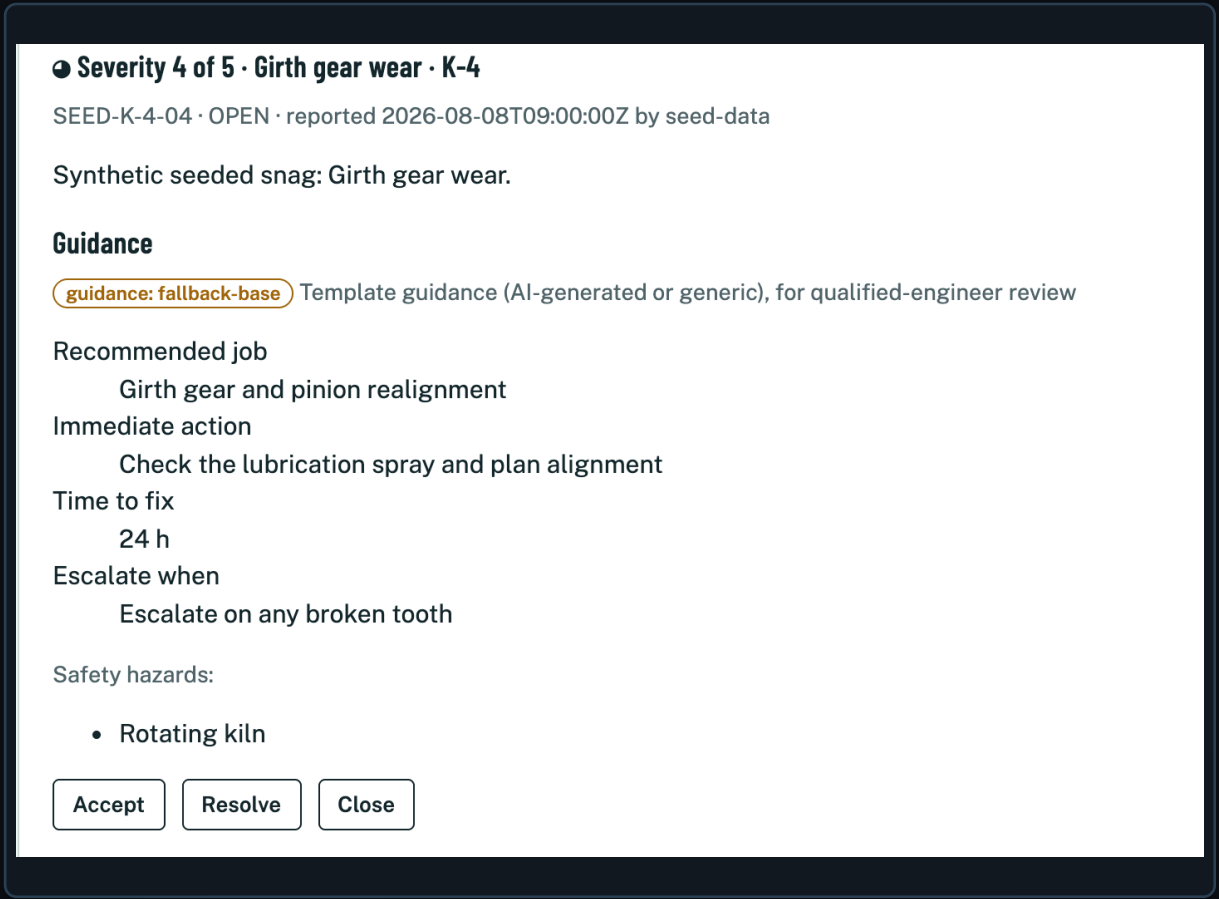
3 Privacy by default

Emails and phone numbers are stripped from free text before storage.

Industry → Model → **Template** → Snag form → Guidance

Screenshot of the local prototype with Gemini not configured: the generic fallback template is shown and labelled. Sample catalogue data.

Step 2 | Resolve: worst snags first, with guidance for the fix



1 Severity-first queue

Each card: recommended job, immediate action, time to fix, hazards, escalation.

2 Close recomputes health

Closing replays history and updates score, lifespan and pollution, plus an audit entry.

3 Honest about access

No auth in the demo. The role switch guides the workflow; it is not a security control.

WORKFLOW STATES (EACH TRANSITION IS AN APPEND-ONLY EVENT)



Screenshot of the local prototype. Seeded snags are synthetic. No authentication in the prototype: the role switch is workflow guidance only.

Step 3 | Health, remaining life and pollution now move together

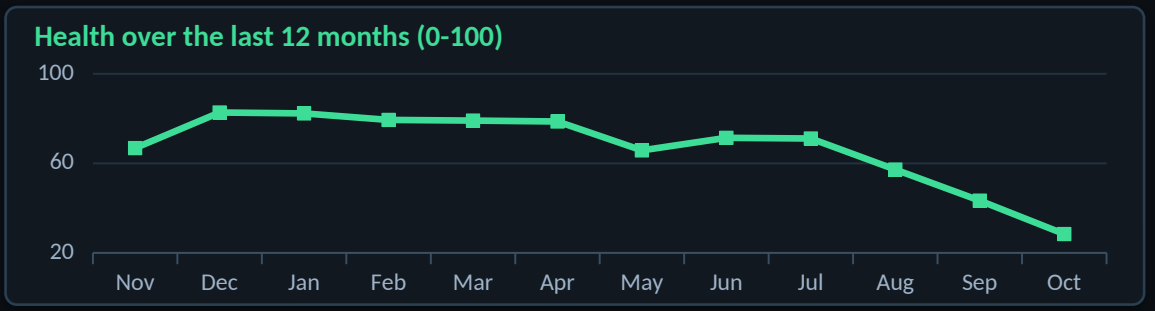
Indicator	Per day now	Reference baseline (healthy machine about +0.5% to +1.5%)	Over remaining lifespan	Source
kg solid waste/day	5575.17	5000	1908628	illustrative assumption
soil-contamination index points/day	13.38	12	4581	illustrative assumption
kg COD/day	28.7	25	9826	illustrative assumption
kg BOD/day	5.74	5	1965	illustrative assumption
m3 effluent/day	918.44	800	314423	illustrative assumption
kg CO2e/day	45549.61	43200	15593643	illustrative assumption
kg NOx/day	113.87	108	38984	illustrative assumption

28.4 / 100

health band: Critical

~342 days

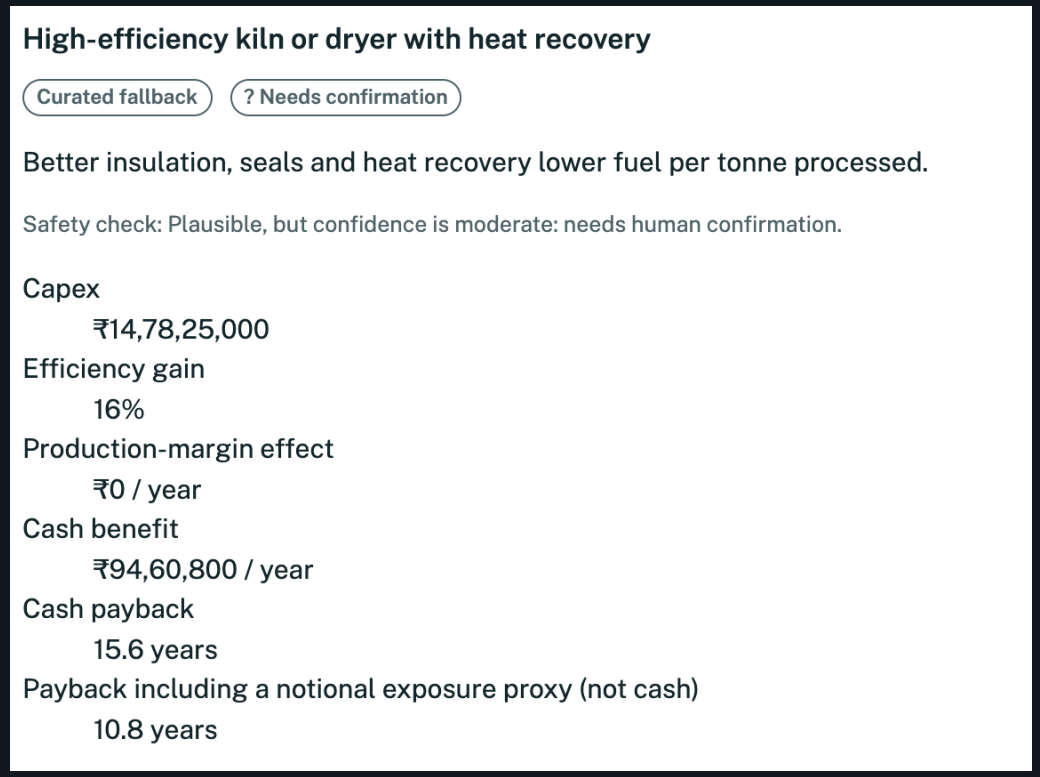
estimated remaining life (range 257-428)



Computed by the engine, never by the LLM

Kiln K-4, synthetic demo fleet. Health, lifespan and intensity are illustrative engine outputs, not measurements or certification.

Step 4 | Greener options with honest paybacks and India incentives



MSME-adjusted capex
₹14,28,25,000

Adjusted cash payback
15.1 years

With exposure proxy
10.4 years

1 **Three kinds of change**

Machine swaps, operating-process changes and assembly-stage changes, curated or Gemini-suggested.

2 **Two paybacks, no spin**

Cash payback and payback with an exposure proxy. Beyond design life is shown, not capped.

3 **MSME or Large**

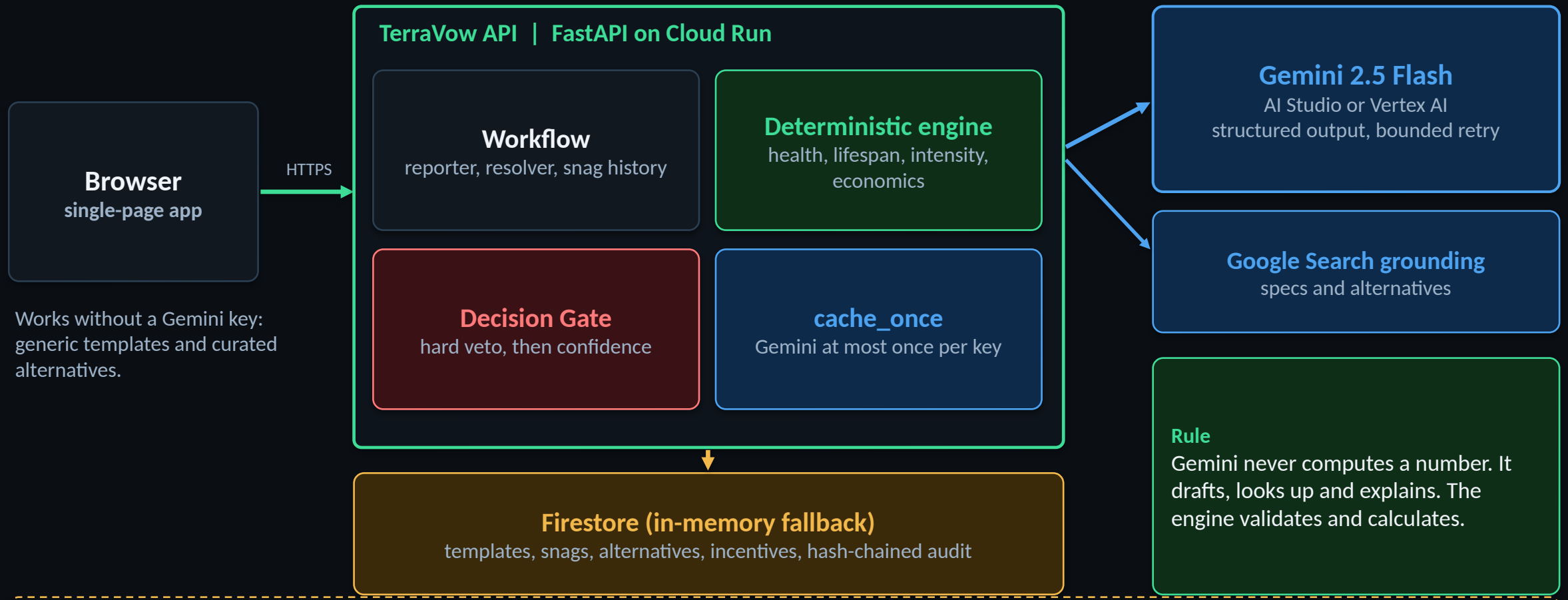
Toggle recomputes incentive-adjusted capex. Real-named schemes are UNVERIFIED; subsidy values are demo-synthetic.

4 **Gate badge on every item**

Needs confirmation (ASK) in the demo: a human decides, nothing auto-approves.

Screenshot of the local prototype (curated fallback alternatives, Gemini not configured). Capex, benefits and incentive values are illustrative; schemes UNVERIFIED.

Architecture: Gemini for language, a deterministic engine for numbers



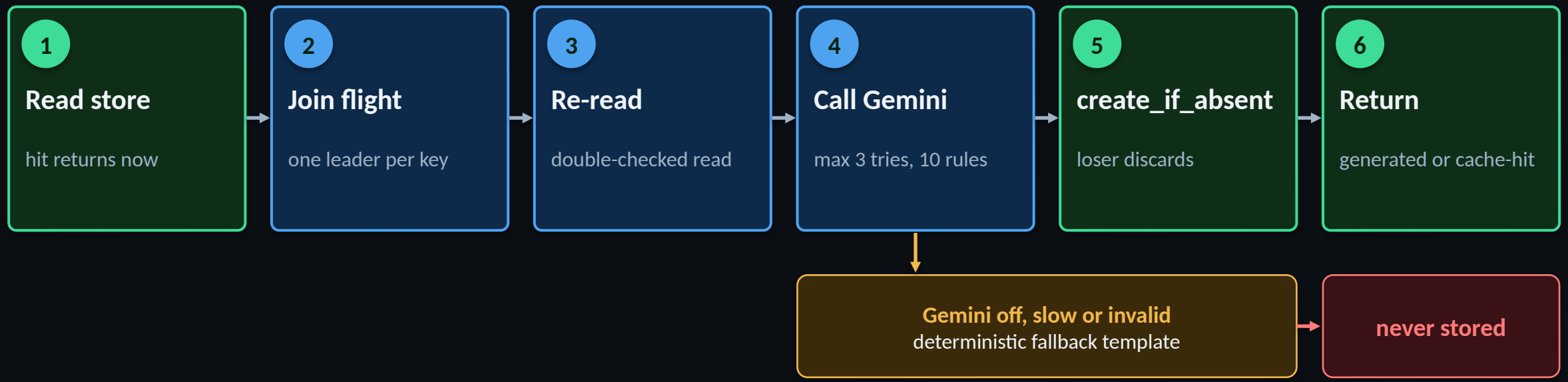
Works without a Gemini key: generic templates and curated alternatives.

Rule
Gemini never computes a number. It drafts, looks up and explains. The engine validates and calculates.

Specified in the full design, not in the demo path: Vertex AI Agent Engine, BigQuery, Vertex AI Search, Pub/Sub, Cloud Storage. Java reference engine runs separately.

Source: docs/system-architecture.md and prototype/README.md. Live Cloud Run and Firestore wiring is checked via /api/health (gemini_mode, store_backend) before submission.

Technical depth: generate once per model, serve everyone



Double-checked read

Never calls Gemini if a result just landed.

Single flight

Waiters share the leader's outcome, even a failure.

Final guard

create_if_absent keeps exactly one document.

Fail safe

Persist error is generated-unsaved; fallback is never cached.

Honest limit: one stored document is guaranteed; the cross-replica lease is not built, so two replicas may both pay for a Gemini call.

Source: docs/system-architecture.md and prototype/app/cache_once.py. Cross-replica lease is specified but not yet built.

Why AI, and why you can trust it: Gemini reasons, the engine calculates

GEMINI REASONS AND WRITES

- Drafts the per-model template
- Looks up specs (Google Search)
- Suggests alternatives as low-high ranges
- Explains results in plain language

VALIDATE

- Schema and 10 template rules
- Bounds and finiteness
- Drop invalid, never clamp
- Midpoint of ranges
- Bounded retry and timeout

THE ENGINE CALCULATES

- Health score and lifespan
- Pollution intensity per domain
- Capex, benefit, payback
- Incentive adjustment
- Gate verdict

No Gemini key? Everything still runs: generic industry template, curated alternatives, the same engine and the same gate. Every fallback is labelled on screen.

Source: docs/system-architecture.md. Gemini 2.5 Flash with structured output, bounded retry (1-5, default 3) and per-attempt timeouts.

Responsible by design: a hard veto that no confidence score can override

1

Hard veto

Total emissions up, or pollution moved air to water?

2

Confidence

0.80 and up: ALLOW | 0.60 and up: ASK | else BLOCK

3

Trust cap

Low data trust can only lower the verdict

4

Audit

Hash-chained entry written before the response

WHAT THE USER SEES

ALLOW Allowed by the safety check

ASK Needs confirmation

BLOCK Blocked by the safety check

AROUND THE GATE

- ✓ Labels: estimate, illustrative, not a certification
- ✓ PII stripped from free text
- ✓ AI options: vendor validation required
- ✓ Demo: nothing reaches ALLOW; a human decides
- ✓ No auth in the demo, stated on screen

Source: prototype/app/gate.py and docs/user-flow.md. Thresholds as built: allow >= 0.80, ask >= 0.60. Outputs are advisory, not certification.

Not another CMMS, ESG ledger or chatbot: TerraVow connects the dots

Capability	Maintenance / CMMS	ESG and carbon	EHS and CEMS	Plain chatbot	TerraVow
Per-machine health from snag history	●	○	○	○	●
Links health to air, water, chemical, soil	◐	○	◐	○	●
Greener option with payback	○	○	○	◐	●
India incentive-adjusted (MSME / large)	○	○	○	○	◐
Auditable, deterministic numbers	●	●	◐	○	●
Veto on fixes that move pollution	○	○	○	○	●

● built ◐ partial or illustrative ○ not offered

Where we are behind: maintenance vendors have large fleets of real sensor data; we have a synthetic demo fleet.

Category-level assessment (our judgement) from docs/competitive-landscape.md; individual products vary. Legend: full = built, partial = limited or illustrative, none = not offered.

Illustrative impact: one Critical kiln, and what each fix would be worth

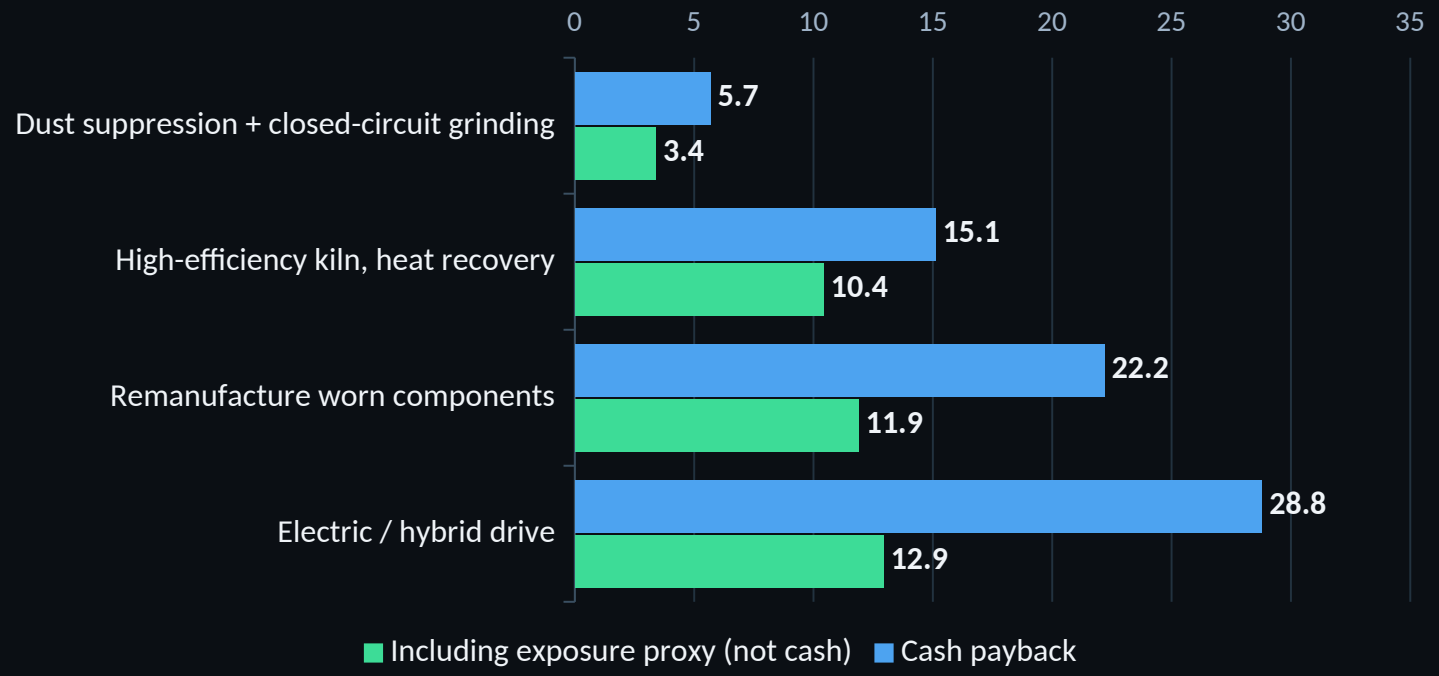
28.4 / 100 health: Critical

~342 days estimated life left

+14.8% effluent vs healthy reference (~118 m3/day)

+5.4% CO2e vs healthy reference (~2.3 t/day)

PAYBACK IN YEARS, MSME, INCENTIVE-ADJUSTED (DESIGN LIFE 25 YEARS)



Dry-stack tailings + water recycling
No payback. Shown, not hidden.

Start with dust suppression
Shortest payback, lowest capex.

All figures: synthetic demo fleet, curated illustrative alternatives, demo-synthetic MSME subsidy, illustrative pollution baselines. Not a measurement, quotation or advice.

Where it lands: India's MSMEs and its highly polluting units

7.47 crore

MSMEs in India (all sectors)

35.4%

of India's manufacturing output is MSME

4,578

highly polluting units reported by SPCBs / PCCs (Dec 2022)

17

CPCB categories of highly polluting industry

BUSINESS MODEL HYPOTHESES | NOT YET VALIDATED

1. **Plant subscription**

   Priced per machine tracked. Reporter and resolver seats included.

2. **Upgrade advisory**

   Turn gated, incentive-adjusted plans into funded projects with partners.

3. **Vendor and financier partnerships**

   Greener-equipment suppliers and green-finance lenders pay for qualified leads.

No market-size or revenue figure is claimed beyond the sourced counts.

Sources: PIB backgrounder 14 May 2026 (MSME counts include services and trade); Rajya Sabha Starred Q101, 15 Dec 2022. Business model is a hypothesis; no revenue claims.

Roadmap: what is built, what is next, and what we have not claimed

DONE

- 4 industries, 20-model sample catalogue
- AI template, cache-once (single process)
- Reporter and Resolver workflow
- Health, lifespan, pollution engine
- Alternatives, incentives, Decision Gate
- Hash-chained audit; 151 tests passing

NEXT

- Cross-replica lease (Firestore)
- Request deadline and waiter cap
- Verify India schemes with authorities
- Real role-based access control
- Measure template depth and latency
- Show grounding citations in the UI

LATER

- Sensor ingestion (Pub/Sub, BigQuery)
- Regulation retrieval (Vertex AI Search)
- Agent Engine orchestration
- Integrity gate: source and baseline checks
- Supply-chain footprint passport
- Calibrate against real machine data

Out of scope for now: official certification and a public registry. Scores are evidence records, not certificates.

Source: prototype/README.md and docs/system-architecture.md. Test result from a local pytest run on 2026-10-18: 151 passed.

Who we are, where to try it, and the 3-minute demo

[Name]	[Name]	[Name]	[Name]
[Role: e.g. AI / backend / product]	[Role: e.g. AI / backend / product]	[Role: e.g. AI / backend / product]	[Role: e.g. AI / backend / product]

Live app	Repo	Demo video	Deck PDF
[Cloud Run URL]	[GitHub URL]	[Video URL]	[PDF link]

DEMO VIDEO STORYBOARD | TARGET 2:50, HARD LIMIT 3:00

0:00	0:15	0:35	1:00	1:20	1:45	2:20	2:40
Hook: the paradox	Pick model, AI template once	Report a snag	Resolve and close	Health and pollution	Options, payback, incentives	Safety gate, architecture	Close and call to action

Say 'illustrative sample image' and 'synthetic demo fleet' on camera. Record the live site, not slides.

TERRAVOW

Fix the machine.
Cut the smoke.
Keep the growth.

Factories will keep growing. TerraVow makes sure the growth is clean.



Try it
[Live URL]

Read it
[Repo URL]

Pilot it
[Contact]

We are looking for: one pilot plant in a highly polluting sector | help verifying India scheme terms | your feedback

Appendix A | Component map, as built

Browser
index.html + vanilla JS

→

FastAPI main.py (app, startup seeding, /api/health) + routes_ext.py (all /api endpoints)

DOMAIN MODULES

industries.py
4 industries, domains, model_key

catalogue.py
search, manual entry

templates.py
validate, form schema, base templates

snags.py
submit, redact, transitions, queue

sustain.py
alternatives, incentives, gate use

engine.py
health, lifespan, intensity, economics

SHARED SERVICES

cache_once.py
Gemini at most once per key

gemini_client.py
template, specs, alternatives

gate.py
hard veto, then confidence

store.py
memory or Firestore, create_if_absent

seed.py
idempotent seeding and reset

EXTERNAL AND STORAGE

Gemini 2.5 Flash + Google Search grounding
AI Studio or Vertex AI; structured output, bounded retry

Storage: Firestore, or in-memory fallback
industries, catalogue, templates, snag_history, snags, alternatives, incentives, seed_meta, audit (hash-chained)

Source: docs/system-architecture.md (Component Map). Simplified into layers for legibility; the full-resolution diagram is in docs/deck/assets/.

Appendix B | Data model: nine collections, one audit chain

industries

key: id

4 industries, their pollution domains and default baselines. Never reset.

catalogue

key: industry--manufacturer--model

20 sample models (5 per industry) plus manual entries. Illustrative economics.

templates

key: model_key--schema version

AI documents only; fallbacks are never stored. Model, time, grounding recorded.

snag_history

key: machine_id

Machine header: tag, install year, model. 6 seeded fleet machines.

snags

key: SNG-12hex

Status OPEN to CLOSED, append-only events, redacted answers. 7 seeded open.

alternatives

key: model_key--country--version

AI suggestions cached per model and country. Curated fallback.

incentives

key: country--scheme_id

9 seeded: 7 real-named UNVERIFIED, 2 demo-synthetic ILLUSTRATIVE.

seed_meta

key: anchor | incentives | complete

Anchor date and seed version. Drives idempotent seeding and reset.

audit

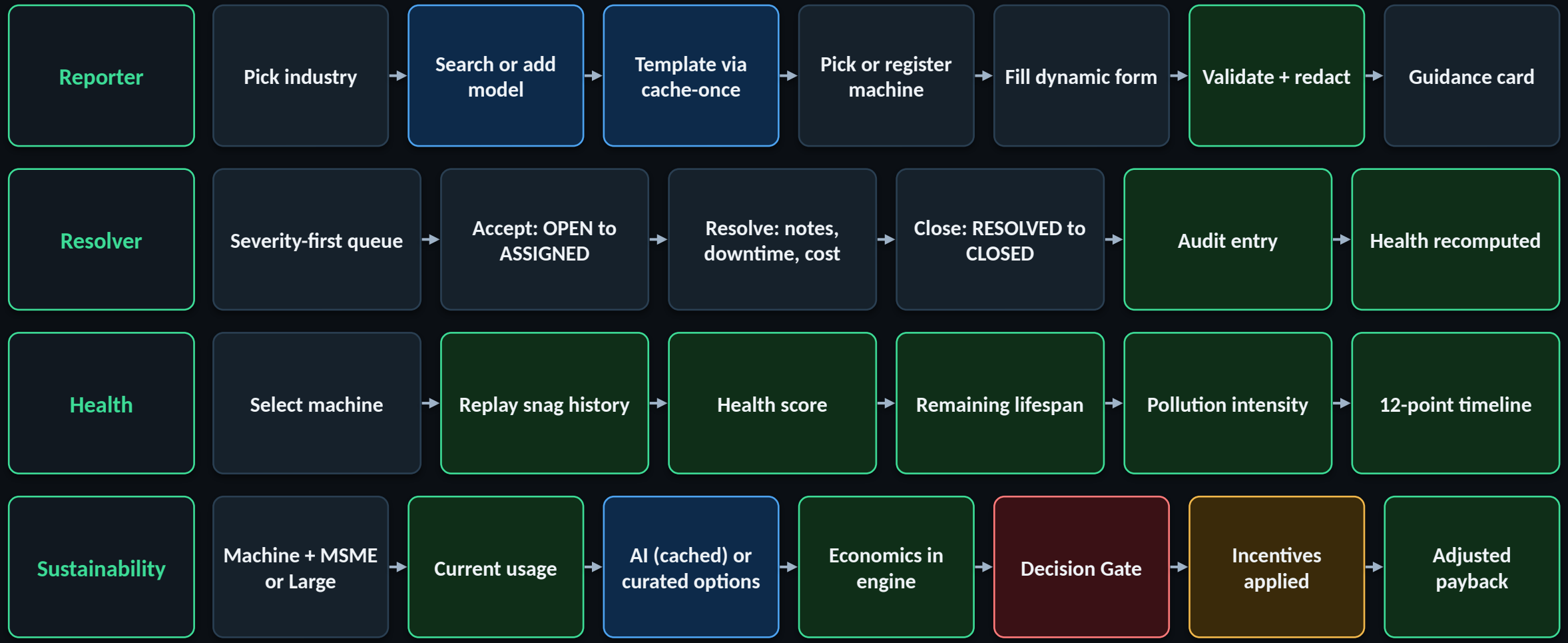
key: AUD-timestamp-hex

Append-only. entry_hash = sha256(prior hash + entry). Kept on reset.

Arrows: foreign-key style links (industry, model_key, machine_id). Full ER diagram: docs/deck/assets/diagram_datamodel.png

Source: docs/system-architecture.md. Catalogue is sample data; seeded snags and incentives are synthetic or illustrative.

Appendix C | The four user flows, step by step



Source: docs/user-flow.md, condensed. Blue = Gemini involved, green = deterministic engine, amber = fallback, red = safety gate. Full flowcharts in docs/deck/assets/.

Appendix D | API surface

Method	Path	Purpose
GET	/api/industries	Four industries and their pollution domains
GET	/api/catalogue/search	Scoped, accent-insensitive search (max 10)
POST	/api/catalogue/manual	Add a model; Gemini + Google Search looks up specs
POST	/api/template/ensure	Cache-once get-or-generate template (Gemini only if absent)
GET	/api/template/{key}	Fetch stored template; never calls Gemini
POST	/api/instances	Register a machine instance
POST	/api/snags	Submit snag: validated, redacted, guidance returned
GET	/api/resolver/queue	Open and assigned snags, severity first
POST	/api/snags/{id}/accept|resolve|close	Role-gated transitions; close recomputes health
GET	/api/machines/{id}/health	Score, lifespan, pollution intensity, 12-point timeline
POST	/api/machines/{id}/recommendations	Alternatives, economics, gate verdicts, incentives
GET	/api/incentives	India schemes by industry and enterprise size
GET	/api/health | /api/registry	Service status | audit hash-chain integrity
POST	/api/demo/reset	Restore seeds, keep audit and cached templates

Source: prototype/README.md. 422 = validation, 403 = wrong role, 409 = invalid transition, 429 = cap reached.

Appendix E | Risks and honest limitations

LIMITATIONS OF THE PROTOTYPE

- Health, intensity, economics use illustrative assumptions; not calibrated on real machines
- Catalogue is 20 sample models; seeded snag histories are synthetic
- India incentives mostly UNVERIFIED; quantified subsidies are demo-synthetic
- No authentication; role switch is workflow guidance
- Cache-once is single-process safe; cross-replica lease not built
- Cached AI alternatives never expire; latency not yet measured live
- Java reference engine is separate from the Python demo path
- Not an official certification; outputs need qualified-engineer review

Incumbents add emissions modules

MITIGATION

Win on the joined-up flow, the safety veto and India incentive logic.

Cold start on real data

MITIGATION

Pilot plants; start from snag logs people already keep.

Over-trust in AI output

MITIGATION

Engine-only numbers, hard veto, labels, human confirmation.

Source: prototype/README.md, docs/product-features.md, docs/competitive-landscape.md.